Lab 3 – Written Justification
Name:Rohith Karning
SRN:PES1UG24AM229
SECTION : D

Second-Hand Product Escrow Marketplace

1. Architectural Choice

We chose Microservices Architecture for the Second-Hand Product Escrow Marketplace. The system is divided into independent services such as Marketplace, Payment & Escrow, Inspection, Dispute Resolution, User Management, Notification, and Audit & Compliance. These services communicate through clearly defined interfaces and can be developed, deployed, and maintained independently.

2. Two Scenario-Related Reasons

1. Independent Scaling:
The marketplace handles different workloads such as product search/listing, order placement, escrow payment, inspection requests, and dispute handling. Microservices allow high-demand services, such as Listing/Catalog or Payment & Escrow, to be scaled independently without scaling the entire system.

2. Fault Isolation and Maintainability:
Payment, inspection, and dispute resolution are separate critical functions. If one service fails, the failure can be isolated rather than bringing down the complete marketplace. Individual services can also be updated or replaced without changing the whole application.

3. Security Advantage

Microservices provide separate security boundaries for sensitive operations. Payment and escrow functions can be isolated and accessed only through defined APIs, while the Audit & Compliance service can record important activities. This improves access control, traceability, and accountability.

4. Performance Benefit

Microservices support independent scaling and parallel processing. Listing, order management, payment, inspection, notification, and audit operations can handle requests independently, reducing unnecessary blocking between functions and improving overall response time and system throughput.

5. Conclusion

The Microservices Architecture is well suited for the Second-Hand Product Escrow Marketplace because it provides **modularity, scalability, fault isolation, security, and improved performance**. Separating important functions such as payment, inspection, dispute resolution, notification, and auditing into independent components makes the system easier to maintain and allows each service to communicate through clearly defined interfaces.